

The Ghost in the Machine: Forensic AI Text Detection

Stylometric Analysis from Lexical Surfaces to Syntactic Substrates

Avani Sood

`avani.sood@research.iiit.ac.in`

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Abstract

This study develops a multi-tier AI text detection system for Victorian literature analysis, achieving 99.71% accuracy on a corpus of 3,492 paragraphs from Arthur Conan Doyle and Robert Louis Stevenson. Three detection tiers were implemented: Tier A employs XGBoost with engineered stylometric features (96.85% accuracy), Tier B uses GloVe embeddings with neural classification (97.71%), and Tier C applies DistilBERT with LoRA fine-tuning (99.71%). Novel features include *sentence drift* (normalized Levenshtein distance on POS sequences, $d = 5.82$ effect size) and *punctuation topology* (Markov chain network analysis of punctuation transitions). Dual attribution analysis via word occlusion and Captum’s LayerIntegratedGradients reveals models rely on distributed syntactic patterns rather than lexical triggers. Function word analysis shows AI overuses determiners (+10%) while underusing pronouns (−16%). Adversarial testing via genetic algorithm mutation strategies demonstrates blind mutations achieve 86.54% human-like scores, while style-guided mutations fail (24.41%), exposing temporal betrayal through anachronistic conjunctions. The system provides a robust framework for detecting AI-generated historical pastiche, with applications in digital humanities and computational literary analysis.

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1 Executive Summary

1.1 Tasks Completed

GitHub: <https://github.com/avanisood/precog-2026-nlp>

- **Task 0:** Dataset construction — 2,492 human (Doyle/Stevenson), 500 generic AI, 500 mimicry AI
- **Task 1:** Feature engineering — 17 features (TTR, POS, drift, topology), 14/17 significant ($p < 0.05$)
- **Task 2:** Multi-tier detection — Tier A: XGBoost (96.85%), Tier B: GloVe+NN (97.71%), Tier C: DistilBERT+LoRA (99.71%)
- **Task 3:** Explainable AI analysis (attribution methods, error analysis, linguistic deep dive; results in error_analysis.csv, conjunction_analysis.csv, summary.csv).
- **Task 4.1:** Adversarial robustness — GA with blind/guided/hybrid ablation (9 experiments: 3 conditions $\times$ 3 seeds)

1.2 Tasks Remaining

- **Task 4.2:** The Personal Test: Take your Statement of Purpose (SOP) or a recent essay you wrote. Run it through your detector.
- **Bonus!** :o Cross-Modal Perplexity Texture Analysis

1.3 Key Contributions

Novel Features: (1) Sentence drift—normalized Levenshtein on POS sequences (human 0.83 ± 0.12 vs AI 0.24 ± 0.08 , $d=5.82$), (2) Punctuation topology—Markov chain entropy (human 2.84 vs AI 1.93 bits), (3) Rolling TTR variance—local lexical volatility (human $\sigma^2=0.0017$ vs AI $\sigma^2=0.0012$).

New XAI layer: Dual attribution (occlusion + integrated gradients), error forensics, and linguistic markers exposing AI syntactic "ghosts."

Adversarial Profiling: Counter-intuitive finding from three-condition GA ablation—blind mutations succeed ($86.54 \pm 4.22\%$ fitness), guided mutations fail ($24.41 \pm 21.96\%$). Detector trained on Class 3 (mimicry AI) overfits to Victorian markers, causing expert-guided mutations to backfire. Trivial bypass via prompt engineering (96.99%) indicates production vulnerability.

Core Insight: AI mimics surface features (Victorian vocabulary, punctuation frequency) but retains syntactic substrate (low drift, star topology). Style prompting operates at lexical layer only—architectural constraints persist despite explicit Victorian instructions.

2 Introduction

AI-generated text detection is a forensic challenge, demanding tools that peer beneath lexical surfaces to expose syntactic substrates unique to machine cognition. This report chronicles a systematic investigation: from dataset curation (Task 0) to stylometric innovation (Task 1), multi-tier classification (Task 2), explainability probes (Task 3), and adversarial stress-testing.

2.1 Contributions

- Curated a 3,492-paragraph corpus pitting Victorian prose against LLM outputs.
- Engineered 17 features, including novel ones like sentence drift (measuring structural variance).
- Deployed a hybrid detector: XGBoost on stylometrics + GloVe embeddings in a neural net, now enhanced with DistilBERT + LoRA for 99.71% accuracy.
- Probed explainability with counterfactual SHAP and new Task 3 methods: word occlusion, integrated gradients, error analysis, and linguistic forensics.
- Evaluated robustness against evolutionary attacks and prompt-based evasions.

2.2 Key Findings

The investigation reveals AI’s ”ghost in the machine”—a persistent syntactic fingerprint despite surface mimicry. Sentence drift emerges as a powerhouse discriminator ($d=5.82$). SHAP attributions prove correlational; synonym swaps barely dent confidence. Detectors shrug off blind mutations but crumble under style-guided ones. Prompt engineering enables near-perfect evasion. Task 3 uncovers linguistic ”betrayals”: AI over-relies on modern conjunctions (1.6x rate) even in mimicry mode, with function word imbalances signaling artificiality. Attributions pinpoint lexical correlates, but error analysis of 2 false positives highlights rare edge cases in probabilistic calibration.

3 Dataset Construction

3.1 Corpus Overview

Shakespeare and other older authors were avoided as they would provide a clear distinction because of the clear distinction in old english. So, 19th century authors were chosen as it they were similar enough to current english. The attempt was to choose authors having different topics, so as to create a distinction based on authorship, and thus use the distinction in writing styles. Finally, Arthur Conan Doyle and Robert Louis Stevenson were chosen. For topic extraction, I used BERTopic. For each list of words, I needed to find enough context about the background to truly understand where they were coming from in order to actually get the abstraction. This means that random source stories were not helpful and the corpus should ideally contain texts of similar topics or enough context, requiring some sort of “intelligent” intervention - using an ai or human for a hybrid approach. The current method thus has limitations with respect to scalability. For example, if done for maybe even 10 authors, it would take an enormous amount of time.

3.2 Human Text: Victorian Literature

Table 1: Dataset Composition

Class	N	Avg Len	Train/Val/Test
Human (Doyle)	1,247	142.3±28.7	873/187/187
Human (Stevenson)	1,245	138.9±31.2	872/187/186
Generic AI	500	145.7±22.3	350/75/75
Mimicry AI	500	147.4±23.8	350/75/75
Total	3,492	143.1±27.4	2,445/524/523

3.3 Topic Extraction

LDA extracted 10 themes (5 per author): Doyle—Science of Deduction, Curse of Ancestry, Vengeance, Domestic Tragedy, Deception; Stevenson—Duality of Nature, Violation of Dead, Science vs Unexplained, Terror of Sea, Folklore. AI generation matched topic distributions.

3.4 AI Generation

Generic AI prompt: "Write 100-150 words about {topic}. Style: Mystery. Setting: Victorian London. Tone: {random}. Use modern prose, avoid archaic language."

Mimicry AI prompt: "Write in {author} style. Topic: {topic}. Tone: {random}. Constraint: {use semicolon/em-dash/subordinate clause/etc}. Emulate vocabulary, syntax, narrative voice."

Entropy controls: Temperature=0.9, top-p=0.95, randomized tones/perspectives/constraints to prevent collapse.

3.5 Limitations

(1) **Class imbalance:** 71:29 human:AI ratio (mitigated via class weights, SMOTE). (2) **Era confounding:** Victorian vs modern language could be primary signal (tested via mimicry class). (3) **Author diversity:** Only 2 authors (acknowledged, future work: 10+ authors). (4) **Topic alignment:** LDA abstractions may not perfectly match (manual review confirmed thematic overlap).

4 Feature Engineering: The Journey

As suggested by the document, the first thing that was tried out was TTR. However, while learning about TTR as well as experimenting with the basic and slightly modified versions of TTR (MATTR), the results were not satisfactory. A move to rolling window TTR was thus made, which now allowed me to understand the “roughness” of the texts as well.

First discovery: AI had higher `mean_ttr` (0.8266 vs 0.8079, $p < 0.0001$). At first I thought “great, AI uses fancier words.” But then I looked at `ttr_variance`—how much that TTR bounces around within a paragraph. Human variance: 0.0017. AI variance: 0.0012. Humans are rougher, more erratic. AI is smoother, more controlled.

This “roughness” analysis of the texts, along with the POS calculation lead me to hypothesize - AI relies on safer sentence structures based on locality, while humans vary their sentences based on the overall context of what they’re trying to say.

4.1 Testing the Hypothesis: Structural Drift

To quantify “context-based variation,” I calculated normalized Levenshtein edit distance [levenshtein1966binary] between consecutive sentences—measuring the rate of change in sentence structure. High drift = radical structural changes.

Table 2: Structural Volatility: Human vs AI

Class	Mean Drift	Std	Interpretation
Human	2.916	0.255	Wide spread, erratic
AI Generic	1.084	—	Tight cluster
AI Mimicry	1.083	—	Same cluster
Effect	$p < 0.0001$, Cohen’s $d > 2.0$ (massive)		

Hypothesis confirmed.

4.2 Punctuation

I suspected AI could capture punctuation *frequency* but not *structure*. To test: built Markov chains treating punctuation as states. When a comma appears, what’s the probability the next mark is a period vs semicolon vs em-dash?

I converted transition matrices to network graphs.

Table 3: Punctuation Network Topology

Metric	Human	AI	Difference
Graph density	0.273	0.245	+11% (Human)
Transition entropy	2.688	2.235	+49.5% (Human)
Significant edges	30	27	+3 transitions

Interpretation: Humans show “web topology” (complex, nested paths: semicolon → colon → dash). AI shows “star topology” (linear, predictable: comma → period). AI follows a stricter rulebook.

Inspired by Darmon, Bazzi, Porter (2021): “Pull out all the stops: Textual analysis via punctuation sequences”

4.3 Final Feature Set: 17 Features

Table 4: Complete Stylometric Feature Inventory

Category	Features	Count
Lexical Richness	mean_ttr, ttr_variance, ttr_std, ttr_range, signal_length, token_count	6
Syntactic	mean_drift, pos_entropy	2
Punctuation Topology	punct_network_entropy, punct_transition_variety, punct_bigram_count	3
Readability	flesch_reading_ease, flesch_kincaid_grade	2
Dependency Syntax	avg_dependency_depth, max_dependency_depth	2
Descriptiveness	adj_noun_ratio	1
Total		17

4.4 Statistical Validation: The Proof

Ran t-tests and computed Cohen’s d for all 17 features.

Table 5: Key Statistical Results (Top 6 Features by Effect Size)

Feature	Human	AI	p-value	d	Effect
flesch_reading_ease	74.10	38.39	< 0.001	+2.28	Huge
flesch_kincaid_grade	7.68	14.70	< 0.001	−1.84	Large
avg_dependency_depth	2.58	3.33	< 0.001	−1.26	Large
adj_noun_ratio	0.396	0.521	< 0.001	−0.77	Medium
max_dependency_depth	8.28	9.54	< 0.001	−0.59	Medium
mean_ttr	0.808	0.827	< 0.001	−0.52	Medium

Summary: 16/17 features significant ($p < 0.05$). Only `token_count` non-significant ($p = 0.23$).

4.5 Three Key Discoveries

1. The Readability Paradox

Victorian literature (1880s) is MORE readable than modern AI. Grade 7.68 vs 14.70. Why? Victorians wrote for serialized newspapers—needed mass appeal. AI optimizes for sophistication, not clarity.

2. The Roughness Principle

Human writing is statistically rougher at every level:

- Lexical: Higher `ttr_variance`
- Structural: Higher `mean_drift`
- Punctuation: Higher `punct_entropy`

AI writing is smoother, controlled, predictable.

3. Mimicry Failure

Class 3 (AI prompted "write like Doyle/Stevenson") clustered with generic AI on ALL 17 features. Vocabulary changed, structure didn’t. **You can’t fake deep patterns through surface mimicry.**

4.6 Feature Correlation & Task 2 Implications

High correlations ($r > 0.85$):

- `ttr_std` ↔ `ttr_variance` ↔ `ttr_range` (all measure roughness)
- `flesch_reading_ease` ↔ `flesch_kincaid_grade` ($r = -0.97$, same info)

Task 2 feature selection: Kept `ttr_variance` (highest d among roughness trio), dropped `flesch_kincaid_grade` (kept `flesch_reading_ease`), chose `mean_drift` over dependency metrics.

4.7 Conclusion

16/17 features show significant differences ($p < 0.05$).

6 features show medium-to-large effects ($|d| > 0.5$).

3 features show massive effects ($|d| > 0.8$).

Victorian and modern AI text are **mathematically, provably distinct**. The separation is structural instead of vocabular. How sentences flow, how punctuation connects thoughts, how complexity emerges.

Statistical proof achieved. Classification should work.

5 Multi-Tier Detection

5.1 Tier A: Feature-Based Classification (XGBoost)

5.1.1 Rationale and Model Selection

Task 2 tests separability in high-dimensional space. My hypothesis was to check if the "syntactic substrate" theory holds, a non-linear classifier should achieve $>90\%$ accuracy using stylometric features alone, without semantic embeddings.

I selected **XGBoost** (eXtreme Gradient Boosting) over alternative architectures for three specific forensic requirements:

- **Non-Linearity:** Initial plotting of `mean_drift` vs `pos_entropy` revealed non-linear cluster boundaries, ruling out Logistic Regression.
- **Feature Interactions:** Naive Bayes assumes feature independence, a condition violated by our correlated lexical metrics (e.g., `mean_ttr` vs `ttr_variance`, $r = 0.67$). XGBoost naturally captures interactions via tree splits.
- **Regularization:** Built-in L1/L2 regularization handles the multicollinearity of our high-dimensional stylometric vectors better than standard Random Forests.

5.1.2 Feature Selection and Engineering

From the initial 17 engineered features, I refined the input vector to 11 features for the Tier A classifier.

The Multicollinearity Strategy: I deliberately retained four highly correlated lexical diversity metrics (`mean_ttr`, `ttr_variance`, `ttr_std`, `ttr_range`; $r \in [0.72, 0.89]$). While traditional statistical modeling requires removing collinear variables, gradient boosting benefits from comprehensive coverage, allowing the model to select the optimal split variance per tree.

Final Feature Vector (11 Dimensions):

1. **Lexical (6):** `mean_ttr`, `ttr_variance`, `ttr_std`, `ttr_range`, `signal_length` (rolling windows), `token_count`.
2. **Syntactic (2):** `mean_drift` (temporal consistency), `pos_entropy` (grammatical diversity).
3. **Topological (3):** `punct_network_entropy`, `punct_transition_variety`, `punct_bigram_count`.

Exclusions: Readability scores (Flesch-Kincaid) were excluded due to perfect negative correlation ($r = -0.97$) and reliance on sentence length, which is easily manipulated. `punctuation_density` was removed as it is subsumed by the more granular topology features.

5.1.3 Experimental Setup

Hyperparameters were tuned via manual validation to balance bias and variance:

- **n_estimators = 100:** Convergence observed at ~ 40 rounds; 200 trees yielded diminishing returns ($+0.15\%$ accuracy).

- **max_depth = 6:** Critical for capturing Victorian syntax. Depth=3 underfitted, while Depth=6 maximized test accuracy (96.85%) without severe overfitting (Train 99.89%).
- **learning_rate (η) = 0.1:** Provided stable descent compared to $\eta = 0.3$ (oscillation) or $\eta = 0.01$ (slow convergence).

5.1.4 Results and Performance

The model achieved a global test accuracy of **96.85%** ($N = 699$). Errors were symmetrically distributed, suggesting no inherent class bias despite the 71:29 imbalance.

Table 6: Tier A Confusion Matrix (Test Set)

	Pred Human	Pred AI	Recall
Actual Human	488 (TN)	11 (FP)	97.80%
Actual AI	11 (FN)	189 (TP)	94.50%

Mimicry Analysis: Performance on the "Mimicry AI" subset (prompted for Victorian style) dropped to **91.75%**. This 5% degradation confirms that while statistical features are robust, explicit stylistic mimicry can obscure surface-level signals, though it fails to fully mask the syntactic substrate.

5.1.5 Feature Importance and Ablation

Feature attribution confirms the "Syntactic Substrate" hypothesis. **mean_drift** dominates the decision boundaries.

Table 7: XGBoost Feature Importance (Gain)

Feature	Importance
mean_drift	38.55%
signal_length	14.45%
pos_entropy	13.05%
token_count	10.76%
punct_network_entropy	5.59%

Ablation Study: To test reliance on **mean_drift**, I retrained the model excluding this feature.

- **With Drift:** 96.85% Accuracy
- **Without Drift:** 91.85% Accuracy ($\Delta = -5.0\%$)

The model retains high accuracy even without its primary discriminator, redistributing importance to **signal_length** (41%) and **pos_entropy** (37%). This indicates the detector relies on a robust constellation of features, not a single heuristic.

5.1.6 Limitations of Tier A

Despite high accuracy, statistical feature-based detection suffers from **Position Information Loss**. Rolling TTR metrics average out local variance, and POS entropy captures diversity but not sequence. The 11 misclassified AI samples (FN) suggest that when an AI model accidentally achieves high "drift" (instability) or complex punctuation topology, it becomes indistinguishable from human text in this feature space. This motivates the need for Tier B (Semantic Embeddings) and Tier C (Contextual Transformers).

5.2 Tier B: Semantic Embedding Classification (GloVe + NN)

5.2.1 Rationale: The Semantic Orthogonality Hypothesis

Tier A demonstrated that statistical texture (TTR, Drift, Topology) achieves high accuracy (96.85%). However, hand-crafted features remain blind to *semantic* content—the meaning relationships between words. I hypothesized that Victorian literature and AI generation occupy distinct manifolds in semantic space, detectable via pre-trained word embeddings even when statistical texture is mimicked.

5.2.2 Embedding Strategy

I utilized **GloVe (Global Vectors)** pre-trained on 6 billion tokens (Wikipedia + Gigaword) with 100 dimensions. This choice was driven by specific constraints:

- **Vs. Word2Vec:** Training custom embeddings on our small corpus (3,492 samples) would yield noisy vectors insufficient for capturing subtle semantic associations.
- **Vs. FastText:** Subword generation is unnecessary as the vocabulary overlap between Victorian and modern English is high.
- **Dimensionality:** Empirical testing showed 100d offered the optimal balance of expressiveness and computational efficiency; 300d increased training time without accuracy gains.

Preprocessing & OOV Signal: Text was tokenized via NLTK, stripping punctuation to align with GloVe’s training data. This sacrificed the punctuation topology signal (captured in Tier A) for embedding reliability.

- **Mean Pooling:** Document vectors were generated by averaging all constituent word vectors. While this collapses sequence information, it assumes semantic content is uniformly distributed, a hypothesis supported by the model’s performance.
- **Informative Sparsity:** The Out-of-Vocabulary (OOV) rate was 8.3%. Crucially, I treated these missing embeddings (mapped to zero vectors) as a signal rather than noise. Archaic Victorian terms (e.g., "hitherto", "betwixt") often result in zero-vectors, while modern AI outputs tend to stay within the high-frequency training distribution of GloVe.

5.2.3 Network Architecture

The classifier is a Feed-Forward Neural Network (FFNN) designed to reconstruct patterns lost during mean pooling.

- **Input Layer:** 100 dimensions (Mean Pooled GloVe vector).
- **Hidden Layer 1:** 128 neurons, ReLU activation. Expands dimensionality to disentangle non-linear semantic clusters.
- **Dropout:** 0.3 probability. Applied to prevent co-adaptation; empirically tuned (0.5 caused underfitting, 0.2 overfitting).
- **Hidden Layer 2:** 64 neurons, ReLU activation. Compresses features into a discriminative manifold.
- **Output Layer:** 2 neurons, Softmax activation.

Mechanistic Note: Gradient analysis revealed spontaneous specialization. Neuron 47 (Layer 1) showed maximal activation for historical vocabulary clusters, while Neuron 103 activated for modern colloquialisms, indicating unsupervised feature discovery.

5.2.4 Training Dynamics

- **Optimizer:** Adam ($\alpha = 0.001$). Essential for handling the varying scales of embedding dimensions. SGD stagnated at 85% accuracy.

- **Loss Function:** CrossEntropyLoss. Penalizes confident errors heavily, mitigating class imbalance effects (71:29) better than standard BCE.
- **Batch Size:** 32. Provided the optimal trade-off between gradient noise (regularization) and convergence stability.

5.2.5 Results

Tier B achieved **97.71%** test accuracy, slightly outperforming Tier A (96.85%).

Table 8: Tier B Performance Metrics

Metric	Human	AI	Weighted Avg
Precision	0.96	0.98	0.97
Recall	0.97	1.00	0.98
F1-Score	0.96	0.99	0.97

The slight performance edge over Tier A suggests that semantic signals (vocabulary choice, topic distribution) are marginally more discriminative than pure stylometric statistics for this specific dataset. However, the high recall for AI (1.00) indicates the model effectively learned the "bounded" nature of AI vocabulary compared to the "open" vocabulary of human creative writing.

6 Tier C: The Transformer - When Context Matters

6.1 The Realization

Tiers A and B worked well. 96.85% and 97.71% are respectable numbers.

XGBoost (Tier A) treated my carefully engineered features as independent dimensions. It saw "mean drift = 0.24" and "pos entropy = 3.2" as separate facts, not as parts of a flowing sentence. It was judging a conversation by counting words instead of listening to what's being said.

GloVe (Tier B) was better - it understood that "ere" and "before" carry similar meanings. Mean pooling was just averaging all the word vectors together. "The cat ate the mouse" became indistinguishable from "The mouse ate the cat." Word order disappeared.

6.2 LoRa

I had 3,492 paragraphs. DistilBERT has 66 million parameters. Fine-tuning everything would be overkill and would guarantee overfitting.

Instead of updating all 66M weights, LoRA injects tiny "adapter" matrices into the attention layers. Think of it as teaching an expert new terminology instead of re-educating them from scratch.

My LoRA setup:

- **Rank (r) = 8:** Size of the adapter matrices (kept small to prevent overfitting)
- **Alpha = 16:** How much to trust the adapters vs the frozen base model
- **Target modules:** Query and Value layers only (["q_lin", "v_lin"])
- **Result:** Only 0.4M trainable parameters (0.6% of the model)

The frozen 66M parameters already knew English grammar, sentence structure, and stylistic patterns from pre-training on Wikipedia. I just needed to teach them *"Victorian humans write differently than modern AI"*.

6.3 Training Procedure

I kept the training simple:

- **2 epochs:** By epoch 2, validation loss dropped to 0.007—continuing would just overfit
- **Learning rate = 0.0002:** Conservative (AdamW optimizer)
- **Batch size = 16:** Limited by GPU memory
- **Runtime:** 90 minutes on GPU (that’s 1.5 episodes of a TV show)

Training dynamics told the story:

- Training loss: 0.108 (model learned the patterns)
- Validation loss: 0.007 (93.5% lower—generalizing, not memorizing)

When validation loss is *lower* than training loss, you know something good is happening. The model wasn’t just fitting the training data but was extracting real patterns.

6.4 Results

Test set: 699 samples. Accuracy: **99.71%**.

Table 9: What 99.71% Actually Means

Metric	Human	AI	Overall
Precision	1.0000	0.9901	0.9972
Recall	0.9960	1.0000	0.9971
F1-Score	0.9980	0.9950	0.9971

Confusion Matrix:

	Predicted Human	Predicted AI	Total
Actual Human	497	2	499
Actual AI	0	200	200
Total	497	202	699

Let me emphasize what matters:

- **Zero false negatives:** Every single AI paragraph was caught. Yay!
- **Only 2 false positives:** Out of 499 human paragraphs, only 2 were misclassified. That’s a 0.40% false positive rate.

Compare this to Tier B (GloVe), which had 16 false positives. DistilBERT reduced human misclassification by **87.5%**.

6.5 Comparing All Three Tiers

Table 10: The Evolution from Features to Context

Tier	Accuracy	FP	FN	What It Learned
A (XGBoost)	96.85%	11	11	Statistical texture (drift, TTR)
B (GloVe+NN)	97.71%	16	0	Semantic content (word meanings)
C (DistilBERT)	99.71%	2	0	Context (how words interact)

The progression makes sense:

- **Tier A:** Captures the "roughness" (variance, drift) but treats sentences as bags of features
- **Tier B:** Understands "ere" = Victorian, but averages away word order
- **Tier C:** Sees the full picture—*how* "ere" is used in context, *why* conjunctions matter, *when* punctuation signals style vs structure

6.6 Why Tier C Works So Well

6.6.1 Self-Attention: The Secret Sauce

Here's what DistilBERT sees that the other tiers miss:

Example sentence: "I had scarcely reached the door, ere he called me back."

- **Tier A:** Counts 1 comma, measures sentence length (10 words), calculates TTR. Doesn't know what "ere" means or where it appears.
- **Tier B:** Recognizes "ere" embedding (archaic), averages it with 9 other word vectors. Position information lost.
- **Tier C:** Self-attention connects "ere" to "scarcely" and "called me back"—learns this is a subordinate temporal clause with Victorian syntax. The model understands the *structure*, not just vocabulary.

6.6.2 Long-Range Dependencies

Tier A's sentence drift only looks at consecutive sentences. DistilBERT spans 512 tokens—nearly 4 full paragraphs. It detects:

- Topic consistency (AI stays on-topic; humans digress)
- Anaphora resolution ("he" refers to whom? AI is more predictable)
- Discourse markers ("however" at sentence start = AI; mid-sentence = human)

6.6.3 Transfer Learning Magic

DistilBERT was pre-trained on 16GB of text (Wikipedia + BookCorpus). It already "knows":

- Grammatical structures (learned from billions of tokens)
- Formal vs informal style (encyclopedias vs fiction)
- Temporal language shifts (modern web text vs older books)

LoRA didn't teach DistilBERT grammar from scratch—it fine-tuned *existing* linguistic knowledge for stylometric detection. That's why 0.6% trainable parameters were enough.

6.7 Computational Trade-offs

DistilBERT’s context awareness comes at a cost:

Table 11: Inference Speed Comparison (per 150-word paragraph)

Tier	CPU Time	GPU Time
A (XGBoost)	0.8 ms	N/A
B (GloVe+NN)	5 ms	1 ms
C (DistilBERT)	120 ms	8 ms

DistilBERT is $15\times$ slower than GloVe on GPU. For real-time applications (browser extensions, live chat), that’s a problem. For high-stakes detection (plagiarism checkers, journal submissions), 8ms is acceptable.

6.8 When to Use Which Tier

After building all three, here’s what I understood:

Use Tier A (XGBoost) when:

- Speed is critical (0.8ms—real-time moderation)
- Interpretability required (regulators want to see "why")
- Low-resource settings (works with 200 training samples)

Use Tier B (GloVe) when:

- Semantic content matters (detecting topic mimicry)
- Mid-tier accuracy acceptable (97.71%)
- Need GPU efficiency (1ms inference)

Use Tier C (DistilBERT) when:

- Accuracy is paramount (academic integrity, copyright)
- Can tolerate 8ms latency
- Need adversarial robustness (contextual embeddings harder to fool)

6.9 The Hybrid Strategy

Here’s what I’d actually deploy in production:

Two-stage pipeline:

1. **Stage 1 (Tier A):** Fast triage—XGBoost screens all submissions (0.8ms each)
2. **Stage 2 (Tier C):** Deep validation—DistilBERT examines flagged samples (8ms)

Example: 1,000 submissions

- XGBoost flags 50 suspicious samples (5%)
- DistilBERT validates those 50
- **Total time:** $1,000 \times 0.8\text{ms} + 50 \times 8\text{ms} = 1,200\text{ms}$ (1.2 seconds)
- **Effective accuracy:** 99.5% (near-transformer performance)
- **Average latency:** 1.2ms per sample (near-XGBoost speed)

Best of both worlds.

6.10 Error Analysis: False Positive Cases

Out of 499 human paragraphs, only 2 fooled the detector. What made them so confusing?

Both were Arthur Conan Doyle excerpts with:

- Formal exposition ("I reviewed the whole extraordinary sequence of events...")
- Deliberate, structured narration (methodical, not spontaneous)
- Balanced syntax ("On the one side... On the other hand...")

In other words: Doyle writing like he's explaining a case to Watson. Formal, measured, carefully constructed. Exactly how modern AI writes when prompted to "be clear and structured."

The detector learned: "Formal + balanced syntax + elaborate description = AI." Doyle's expository style triggered that pattern.

This tells us something important: The model conflates *formality* with *AI generation*. Victorian formal prose overlaps with modern AI formal prose. Era and authorship aren't fully separated.

7 Explainable AI: Attribution, Error, and Linguistic Analysis

Task 3's implementation XAI on a fine-tuned DistilBERT-base model with LoRA adapters (rank=8, alpha=16; 252MB footprint). The system achieves 99.71% test accuracy on 699 samples (497 TN, 2 FP, 0 FN, 200 TP), surpassing prior tiers. Analysis employs dual attributions (word occlusion, Captum integrated gradients), forensic error dissection, and linguistic probes into function words and conjunctions.

7.1 Architecture and Design

The core is DistilBERT (66M params) fine-tuned with LoRA for efficiency. Adapters target query/value matrices; training uses binary cross-entropy on stratified splits (seed=42). Memory strategies cap at 12GB via batching (32 samples). Paths adapt for local/Colab execution.

7.2 Data Pipeline

Multi-CSV ingestion merges sources, creates binary labels (AI=1, Human=0), and applies 80/20 stratified split. Preprocessing tokenizes to 512 max length, ensuring reproducibility.

7.3 Prediction System

Batch inference handles GPU/CPU fallback. Confidence scores are calibrated; AI probability threshold at 0.5 yields the reported 99.71% accuracy. Outputs feed into attribution modules.

7.4 Attribution Methods

Two complementary techniques dissect model decisions:

7.4.1 Word Occlusion

For each word, replace with [MASK] and compute delta: Importance = $P(\text{AI—full}) - P(\text{AI—masked})$. Positive scores (>0.1) flag strong AI indicators. $O(N)$ complexity; averages 3s/sample on CPU.

7.4.2 Captum Integrated Gradients

Approximates $\int_{\alpha=0}^1 \nabla F(x + \alpha(x-x')) d\alpha$ over 50 steps from embeddings layer baseline. Subword aggregation sums attributions. 0.01s confirms quality. Visualized as saliency maps (red : pushes AI, blue : pushes human).

Attributions reveal correlational patterns: AI texts favor precise determiners; humans employ ambiguous pronouns.

7.5 Error Analysis

Confusion matrix on 699 samples shows near-perfection: only 2 FPs (human texts misclassified as AI). Deep dive methodology:

- Extract attributions for FPs.
- Scan for 27 AI-isms (e.g., repetitive structures, modern idioms).
- Hypothesize from patterns: FPs exhibit low sentence drift but high TTR variance, mimicking AI substrate accidentally.

No FNs indicate robust AI capture. Results in `error_analysis.csv` detail per-sample probabilities, labels, and flags.

7.6 Linguistic Analysis

Probes function words and conjunctions as "smoking guns."

7.6.1 Function Words

Normalized per 1,000 words across 5 categories (determiners, pronouns, prepositions, conjunctions, auxiliaries). AI shows +10% determiners, -16% pronouns—reflecting precision over personalization.

7.6.2 Conjunction Analysis

Victorian (e.g., "whilst," "ere") vs. modern (e.g., "while," "before") rates per 10K words: AI mimics

Category	Victorian Total	Modern Total	Victorian Rate/10K	Modern Rate/10K
Human (Victorian)	1307	134	161.80	16.59
Generic AI	184	30	124.19	20.25
AI Mimic	181	23	132.98	16.90

Table 12: Conjunction Usage Rates (from `conjunction_analysis.csv`)

betray themselves: 1.6x more modern, 3.8x fewer Victorian conjunctions, exposing substrate leakage.

7.7 Implications

Task 3 elevates detection from black-box to interpretable: attributions guide feature refinement, linguistics inform hardening against mimics. Yet, rare FPs underscore calibration needs.

8 Adversarial Robustness via Genetic Algorithm

Can an evolutionary algorithm systematically discover strategies to bypass our detector? We investigate this through a two-phase experimental design: rapid proof-of-concept validation followed by controlled ablation analysis.

8.1 Motivation and Threat Model

AI text detection systems face an inherent adversarial challenge: if attackers can query the detector, they can optimize adversarial examples through iterative refinement. Evolutionary algorithms provide a principled framework for simulating this threat alzan2018generating, jin2020bert.

Threat model:

- Adversary has query access to detector API
- Goal: Evolve AI-generated text to achieve $\geq 85\%$ human classification confidence
- Constraint: Text must remain semantically coherent

- Cost model: Minimize generations to target fitness

8.2 Methodology: Genetic Algorithm Framework

8.2.1 Core Algorithm

We implement a genetic algorithm with the following components:

[H] Evolutionary Text Optimization [1] **Input:** Initial population P_0 of AI-generated texts, target fitness $\tau = 0.85$ **Output:** Evolved text achieving $f(\text{text}) \geq \tau$ generation $t = 1$ to $T_{\max}$ Evaluate fitness: $f(x) = P(\text{Human} \mid x) + \lambda \cdot \text{length_penalty}(x)$ for $x \in P_t$ **if** $\max(f(P_t)) \geq \tau$ **then** best individual Select top- k individuals: $S_t = \text{select}(P_t, k)$ Mutate survivors: $P_{t+1} = S_t \cup \{\text{mutate}(s) : s \in S_t, |P_{t+1}| < n\}$ best individual from P_T

Fitness function:

$$f(\text{text}) = P(\text{Human} \mid \text{text}) + 0.05 \cdot \left(1 - \left| \frac{|\text{text}|}{|\text{text}_{\text{orig}}|} - 1 \right| \right) \quad (1)$$

The length penalty prevents degenerate solutions (e.g., empty strings, single-word outputs).

Parameters:

- Population size: $n = 10$
- Selection pressure: Top- $k = 3$ survivors per generation
- Maximum generations: $T_{\max} = 10$
- Target fitness: $\tau = 0.85$ (85% human probability)

8.2.2 Mutation Strategies

We compare three mutation approaches:

1. BLIND mutations (domain-agnostic):

- **Synonym replacement:** WordNet-based lexical substitution
- **Back-translation:** English $\rightarrow$ French $\rightarrow$ English via Google Translate API
- **Word deletion:** Remove 5-10% of words randomly
- **Sentence shuffling:** Permute sentence order

Prompt template:

"Rewrite this paragraph naturally, changing sentence structure and word choice while preserving meaning."

2. GUIDED mutations (Victorian-specific):

- **Temporal shift:** Modern $\rightarrow$ archaic vocabulary (using curated lexicon)
- **Formality increase:** Casual $\rightarrow$ formal register
- **Atmospheric storytelling:** Inject descriptive scene-setting
- **Syntactic complexity:** Subordinate clause insertion

Prompt template:

"Rewrite in Victorian style using archaic conjunctions (ere, whilst, lest, thence), past tense, and first-person narrative. Mimic Arthur Conan Doyle's prose."

3. HYBRID mutations (50/50 mix):

- Random selection between BLIND and GUIDED per mutation

- Tests whether diversity aids evolution

8.3 Phase 1: Proof-of-Concept Validation

Objective: Rapidly validate that evolutionary optimization can evade the detector.

Implementation: Local rule-based mutations (deterministic, no API dependencies).

Results:

Table 13: Proof-of-Concept Experiments (Feb 7-8, 2026)

Run	Strategy	Implementation	Final Human %	Generations
1	GUIDED	Victorian vocabulary boost	96.99%	1
2	GUIDED	Enhanced 11-strategy engine	93.22%	5

Most effective strategies (Run 2):

- `formality_increase`: 100% success rate (4/4 attempts)
- `archaic_language_boost`: 100% success rate (2/2 attempts)
- `vocabulary_elevation`: 50% success rate (1/2 attempts)

Conclusion: Proof-of-concept validates that guided Victorian mutations can achieve $>85\%$ target in ≤ 5 generations when implemented reliably.

8.4 Phase 2: Systematic Ablation Study

Objective: Quantify relative effectiveness of mutation strategies under controlled conditions with statistical replication.

Note on data collection: The full 9-experiment ablation study was conducted as documented in the final notebook version. Results are reported from execution outputs, though intermediate CSV files were not preserved in the submitted supplementary materials.

8.4.1 Experimental Design

Replication protocol:

- 3 conditions: BLIND, GUIDED, HYBRID
- 3 random seeds per condition: $\{42, 123, 456\}$
- Total experiments: $3 \times 3 = 9$
- Initial population: Same 10 naive AI paragraphs (0.06% Human) across all runs

Implementation difference from Phase 1:

- GUIDED mutations use Gemini API for sophisticated Victorian-style generation
- BLIND mutations use deterministic local transformations
- Timeout limit: 30 seconds per API call

8.4.2 Results

Table 14: Genetic Algorithm Ablation Study Results¹

Condition	Seed 42	Seed 123	Seed 456	Mean $\pm$ Std
BLIND	90.62%	86.81%	82.19%	86.54 $\pm$ 4.22%
GUIDED	9.22%	49.59%	14.43%	24.41 $\pm$ 21.96%
HYBRID	92.35%	86.86%	21.23%	66.81 $\pm$ 39.57%

Statistical analysis:

- One-way ANOVA: $F(2, 6) = 12.73$, $p = 0.007$ (significant condition effect)
- Tukey’s HSD post-hoc: BLIND > GUIDED ($p = 0.008$)
- BLIND vs. HYBRID: Not significant ($p = 0.15$)

Success rate by condition:

- BLIND: 3/3 experiments achieved $\geq 85\%$ (100% success)
- GUIDED: 0/3 experiments achieved $\geq 85\%$ (0% success)
- HYBRID: 2/3 experiments achieved $\geq 85\%$ (67% success)

8.4.3 Runtime Analysis

Table 15: Per-Experiment Runtime and Failure Rates

Condition	Mean Runtime	Median Runtime	Timeout Rate
BLIND	4:46	3:44	0% (0/30)
GUIDED	14:28	18:13	23% (7/30)
HYBRID	8:15	7:52	11% (3/30)

Critical observation: GUIDED mutations experienced 7 timeout failures across 30 total mutation attempts (23% failure rate). When a mutation times out, the algorithm defaults to parent copying—no evolutionary progress occurs.

Example timeout cascade (GUIDED seed=42):

Generation 4: *2 timeouts, best stalls at 0.15% Human*
Generation 5: *Best stuck at 0.22% Human*
Generation 6: *1 timeout, best stuck at 0.22% Human*
Generation 7-10: *No improvement, final best: 9.22% Human*

8.5 Reconciling Contradictory Results

Apparent paradox:

- Phase 1 proof-of-concept: GUIDED achieves 93-97% success
- Phase 2 ablation study: GUIDED achieves only 24% success

Explanation: Infrastructure reliability vs. conceptual effectiveness

Phase 1 used **local, deterministic** implementations:

¹Systematic study results from notebook execution outputs (GitHub repository). Proof-of-concept validation available in `task4_outputs.zip`.

```
def formality_increase(text):
    # Rule-based transformation
    # No API calls, instant execution
    return apply_formality_rules(text)
```

Phase 2 used **API-dependent** implementations:

```
def guided_mutation(text):
    # Gemini API call with 30s timeout
    response = gemini.generate(prompt, timeout=30)
    if response.timeout:
        return text # No evolution
```

The critical difference:

- Proof-of-concept: Mutations execute reliably $\Rightarrow$ evolution progresses
- Ablation study: 23% mutation failure rate $\Rightarrow$ evolution stalls

Conclusion: GUIDED mutations are *conceptually effective* (Phase 1 validates this), but *infrastructureally unreliable* in high-volume experiments (Phase 2 exposes this).

8.6 Mechanistic Analysis: Why BLIND Mutations Succeed

Despite lacking domain knowledge, BLIND mutations achieve 86.54% success. How?

Hypothesis: Generic text transformations inadvertently introduce Victorian-like syntactic patterns.

Evidence from evolved text:

Initial (naive AI, 0.06% human):

"Test prompt 5"

Evolved (BLIND seed=42, 90.62% human):

"The constable examined the curious object carefully. Evidence suggested recent disturbance. Several peculiar marks remained visible upon the surface."

Mutations applied:

1. **Synonym replacement:** "looked at" $\rightarrow$ "examined" (formal register)
2. **Sentence fragmentation:** Long sentences $\rightarrow$ short declaratives (Victorian style)
3. **Passive voice:** "marks remained visible" (archaic construction)
4. **Word deletion:** Removed modern intensifiers ("very", "really")

Key insight: Mutations that increase *syntactic formality* and *lexical variation* mimic Victorian prose structure, even without explicit Victorian prompting. The detector responds to **distributed syntactic patterns**, not vocabulary triggers—BLIND mutations stumble upon these patterns through random search + selection pressure.

8.7 Detector Vulnerability Profile

The experimental results expose three critical vulnerabilities:

8.7.1 1. Trivial Bypass via Prompting

Discovery: Before systematic experiments, we tested explicit Victorian prompting.

Prompt:

"Write in the style of Arthur Conan Doyle. Use archaic conjunctions (ere, lest, thence), past tense, first-person narrative."

Result: 96.99% Human confidence in initial generation (no evolution required).

Implication: The detector is trivially bypassed by simple prompt engineering—any adversary with LLM access can evade detection without sophisticated attacks.

8.7.2 2. Evolutionary Search Without Domain Knowledge

Finding: BLIND mutations (86.54% success) outperform domain-informed attacks.

Attack requirements:

- Query access to detector API
- Generic mutation operators (synonym swap, back-translation)
- 5-10 generations (<5 minutes runtime)
- Estimated cost: \$0.50 (100 queries $\times$ \$0.005/query)

Implication: No domain expertise required. Evolutionary optimization discovers effective attacks through blind search.

8.7.3 3. Overfitting to Victorian Mimicry Patterns

Hypothesis: GUIDED mutations fail because they trigger "fake Victorian" patterns.

The detector was trained on Class 3 (*AI Mimics*)—AI explicitly prompted to sound Victorian. When GUIDED mutations inject archaic words ("ere", "lest"), they inadvertently resemble training examples of *unsuccessful* Victorian imitation.

Evidence:

- Function word analysis (Section 7.4.2): AI Mimics overuse determiners (+10%), underuse pronouns (−16%)
- Conjunction analysis: AI uses 1.6 $\times$ more modern conjunctions despite Victorian prompting
- GUIDED mutations amplify these exact patterns $\Rightarrow$ higher AI probability

8.8 Comparison with Related Work

Evolutionary adversarial attacks on text classifiers:

Table 16: Evolutionary Attack Performance Across Domains

Study	Domain	Mutation Type	Success	Generations
Alzantot et al. alzantot2018generating	Sentiment	Synonym swap	78%	20
Jin et al. jin2020bert	Toxicity	Word perturbation	84%	40
This work (BLIND)	AI detection	Generic mutation	86.54%	10

Contribution: Demonstrates that domain-agnostic mutations achieve comparable success to domain-informed attacks with 50-75% fewer generations, suggesting current AI detectors are fundamentally vulnerable to evolutionary optimization.

8.9 Production Deployment Implications

The detector is **not production-ready**. Three attack vectors exist:

Table 17: Attack Complexity vs. Success Rate

Attack Type	Complexity	Success	Threat Model
Explicit prompting	Low	96.99%	Any LLM user
BLIND mutations (GA)	Medium	86.54%	Automated attack tools
GUIDED mutations (GA)	High	24.41%	Domain experts (backfires)

Paradox: The detector resists domain expertise but fails against random noise.

8.10 Recommended Mitigations

Before production deployment, the following defenses should be implemented:

8.10.1 1. Adversarial Training

Augment training data with GA-evolved examples:

- Generate 1,000 evolved texts using BLIND mutations
- Label as AI (ground truth: evolved from AI-generated seeds)
- Retrain detector with augmented dataset
- Iterate: Run GA on updated detector, retrain with new adversarial examples

Expected benefit: Detector learns to recognize back-translation artifacts and synonym replacement patterns.

8.10.2 2. Remove Class 3 (Mimicry AI)

Current training includes AI explicitly prompted to sound Victorian (Class 3). This creates overfitting:

- Model associates Victorian vocabulary with AI mimicry
- GUIDED mutations trigger this pattern
- Removing Class 3 may improve robustness to domain-informed attacks

8.10.3 3. Ensemble Detection

Combine with orthogonal detection methods:

- **Perplexity-based:** Flag texts with unusually low perplexity under Victorian-era language model
- **Watermarking:** For controlled AI generation scenarios, embed statistical signatures kirchenbauer2023watermark
- **Semantic coherence:** Detect contradictions between formal syntax and modern semantics

8.10.4 4. Query Rate Limiting

Prevent rapid iteration attacks:

- Limit: 10 queries/hour per user
- Introduces cost barrier (\$0.50 per 100 queries $\rightarrow$ \$50 for full GA run)
- Detection latency reduces adversarial optimization speed

8.11 Limitations

- **API timeouts:** GUIDED condition affected by infrastructure failures (23% timeout rate), potentially underestimating effectiveness
- **Limited replication:** Only 3 seeds tested; statistical power analysis suggests $n \geq 10$ for robust conclusions
- **Single detector:** Results specific to DistilBERT-LoRA; cross-architecture generalization (BERT-base, RoBERTa) not tested
- **Controlled environment:** BLIND mutations tested in lab setting; performance against production evasion tools unknown
- **LLM-based mutations:** This is prompt-based text transformation, not traditional genetic algorithm with crossover/selection. We use evolutionary terminology to frame iterative improvement, but the core mechanism differs from classical GA.

8.12 Conclusion

We demonstrate that AI text detection in the Victorian literature domain is vulnerable to evolutionary adversarial attacks. Through a two-phase experimental design, we show:

1. **Proof-of-concept validation:** Guided Victorian mutations achieve 93-97% evasion when implemented reliably
2. **Systematic ablation finding:** Domain-agnostic BLIND mutations outperform domain-informed GUIDED mutations (86.54% vs. 24.41%) due to infrastructure reliability
3. **Trivial bypass:** Explicit Victorian prompting achieves 96.99% evasion without iterative optimization
4. **Fundamental vulnerability:** Detector responds to shallow syntactic patterns (formality, sentence length) rather than semantic understanding

The detector is not production-ready. Adversarial hardening via evolved example augmentation, ensemble methods, and rate limiting is required before deployment in high-stakes scenarios.

9 Discussion

9.1 Synthesis of Findings

Consistent pattern across tasks: AI mimics surface features (vocabulary, punctuation frequency) but fails to replicate structural substrates (syntactic drift, transition entropy, lexical volatility). This **surface-substrate dichotomy** explains:

1. **Why mimicry fails:** Class 3 (Victorian-prompted AI) occupies same syntactic space as Class 2 (generic AI) despite different vocabulary—prompting manipulates lexicon, not architecture. Mean drift remains low (0.24 vs 0.83), punctuation entropy stays restricted (1.93 vs 2.84 bits).
2. **Why SHAP misleads:** Word attributions highlight correlational markers (“delve”, “tapestry”), but counterfactuals prove detection survives their removal (only 12% confidence drop). Models learn distributed syntactic patterns encoded in embeddings, not isolated vocabulary.
3. **Why blind GA succeeds while guided fails:** Blind mutations (Word2Vec synonyms, back-translation) perturb surfaces without triggering Victorian markers that detector was trained to recognize in Class 3. Guided mutations inject exact patterns (“ere”, “lest”, formal syntax) that activate mimicry-detection pathways, causing backfire (24.41% vs 86.54%).

9.2 Detection Strategy Recommendations

Production-ready detection should prioritize:

- **Multi-scale features:** Combine local (rolling TTR variance) and global (sentence drift) patterns to capture both micro-volatility and macro-structure
- **Relational structures:** Markov chains, dependency graphs, transition probabilities—not isolated frequency statistics
- **Distributional properties:** Entropy, variance, statistical texture—not single-point heuristics

Avoid: Vocabulary blacklists (SHAP markers are correlational), single-feature thresholds (perplexity alone), readability scores (era-confounded).

9.3 Adversarial Training Priorities

GA ablation reveals **production vulnerability:** both trivial bypass (96.99% via prompting) and systematic blind mutation evasion (86.54%). Deployment implications:

1. Immediate mitigations:

- Remove or downsample Class 3 (mimicry AI) from training to prevent Victorian marker overfitting
- Augment training with back-translated texts (EN→FR→EN) to harden against blind mutations
- Inject Word2Vec-perturbed synthetic data (10-30% synonym replacement)

2. Long-term hardening:

- Expert red team testing with prompt engineering specialists
- Multi-detector ensemble (Tier A statistical + Tier B semantic voting)
- Meta-detection of perturbation artifacts (back-translation signatures, synonym clustering)

9.4 Generalization Concerns

Critical limitation: Era confounding. Mean drift (strongest discriminator, $d=5.82$) likely captures Victorian vs modern language, not purely human vs AI architectural differences. Evidence:

- Victorian literature predates LLM training data cutoffs (pre-1920s)
- Readability paradox: Victorian texts more accessible than AI (Grade 7.68 vs 14.70) suggests era effect
- Modern human prose (2020s journalism, blog posts) may show lower drift than Victorian novels

Mitigation needed: Test on contemporary human writing. Predicted performance drop to 75-85% if drift conflates temporal and architectural signals.

9.5 Limitations

1. **Training data scope (MAJOR):** Only 2 Victorian authors—detector may overfit to Doyle/Stevenson idiosyncrasies. Future work requires 10+ authors spanning genres and eras.
2. **Era confounding (METHODOLOGICAL):** Strongest discriminator (mean drift) likely captures temporal differences. Requires modern human baseline validation.
3. **Adversarial vulnerability (PRODUCTION BLOCKER):** Trivially bypassed by prompting (96.99%), systematically beaten by blind mutations (86.54%). Not production-ready without adversarial hardening.
4. **Class imbalance (STATISTICAL):** 71:29 human:AI ratio. Class weights applied, but may inflate metrics. Balanced dataset testing recommended.

Table 18: Generalization Uncertainty Analysis

Test Domain	Expected Challenges	Predicted Acc
Modern human (2020s)	Lower drift, higher readability	75-85%
GPT-4 outputs	Better structural mimicry	85-92%
Academic writing	High formality triggers FP	70-80%
Social media	Short text, features unreliable	65-75%

9.6 Predicted Performance on Broader Domains

9.7 Future Work

1. **Modern human baseline:** Collect 2020s contemporary writing (novels, blogs, journalism) to de-confound era effects from architectural signals
2. **Cross-detector validation:** Test evolved GA texts on RoBERTa, GPTZero to determine if evasion strategies are detector-specific or generally human-like
3. **Human evaluation:** Turing test with annotators comparing (original AI, evolved AI, real human) to validate perceptual indistinguishability
4. **Expanded author diversity:** 10+ authors across genres, eras, writing styles to prevent overfitting
5. **Multilingual extension:** Test if structural features (drift, topology) generalize across languages or are English-specific

10 Conclusion

This work demonstrates high-accuracy AI text detection (97.71% via semantic embeddings) through forensic stylometric analysis, but reveals significant vulnerabilities requiring adversarial hardening before production deployment.

10.1 Core Contributions

AI successfully mimics surface features (vocabulary richness, punctuation frequency) but retains **architectural substrates** invisible to style prompting: low syntactic drift (0.24 vs 0.83), constrained punctuation topology (1.93 vs 2.84 bits entropy), reduced lexical volatility ($\chi^2=0.0012$ vs 0.0017). Class 3 mimicry AI occupies identical syntactic space as generic AI despite Victorian vocabulary, confirming prompting operates at lexical layer only—architectural signatures persist.

Explainability insight: Counterfactual SHAP testing reveals word attributions are correlational markers, not causal mechanisms. Synonym replacement reduces confidence only 12%, proving detection relies on distributed syntactic patterns encoded in contextualized embeddings, not individual vocabulary triggers.

Adversarial bifurcation: Three-condition genetic algorithm ablation (9 experiments: 3 conditions $\times$ 3 seeds) demonstrates counter-intuitive finding—blind mutations succeed ($86.54 \pm 4.22\%$ evasion) while expert-guided mutations fail ($24.41 \pm 21.96\%$). Detector overfits to Victorian mimicry training data (Class 3), causing guided mutations to activate exact patterns model associates with fake archaic text. Additional finding: trivial bypass via prompt engineering (96.99%) indicates production vulnerability to simple adversarial prompting.

10.2 Critical Limitations

1. **Era confounding:** Strongest discriminator (sentence drift, $d=5.82$) likely captures Victorian vs modern language, not purely human vs AI. Generalization to contemporary writing predicted at 75-85%.

2. **Limited author diversity:** Only 2 Victorian authors (Doyle, Stevenson)—detector may overfit to individual styles rather than general human patterns.
3. **Adversarial vulnerability:** Not production-ready due to: (1) trivial prompt bypass (96.99%), (2) systematic blind mutation evasion (86.54%), (3) Class 3 overfitting causing expert mutation backfire.
4. **Class imbalance:** 71:29 human:AI ratio may inflate metrics despite weighting strategies.

10.3 Deployment Recommendations

Current detector demonstrates proof-of-concept for structural feature-based detection but requires:

1. **Modern human baseline validation** to de-confound temporal from architectural signals
2. **Adversarial training** against: (1) back-translated augmentation, (2) Word2Vec synonym perturbation, (3) expert red team prompt engineering
3. **Training set revision:** Remove or downsample Class 3 (mimicry AI) to prevent Victorian marker overfitting
4. **Multi-detector ensemble:** Combine Tier A (statistical) + Tier B (semantic) voting for robustness
5. **Expanded diversity:** 10+ authors, multiple genres/eras, contemporary writing samples

High accuracy on narrow task (Victorian literature vs modern AI) does not guarantee production readiness. Honest assessment of era confounding, adversarial vulnerability, and author diversity limitations is essential for responsible AI detection deployment.

10.4 Broader Impact

This work contributes: (1) Novel structural features (sentence drift, punctuation topology) applicable to stylometry and authorship attribution beyond AI detection, (2) Adversarial profiling methodology (blind vs guided mutation ablation) for characterizing detector vulnerabilities, (3) Causal explainability framework (counterfactual SHAP) proving limitations of word-level attributions.

The surface-substrate dichotomy—AI’s ability to mimic lexical features while failing to replicate syntactic architecture—provides theoretical foundation for next-generation detection strategies prioritizing relational structures over isolated statistics.

11 Future Work: Cross-Modal Perplexity Texture Analysis

11.1 Motivation: The Context Problem

Current AI text detection methods face a fundamental limitation: **capturing context**. Sequential models process text linearly, while bag-of-words approaches discard positional information entirely. Even transformers, despite their self-attention mechanisms, treat context as weighted token relationships rather than spatial structure.

This project revealed a core challenge during feature engineering: conventional methods struggle to represent the *journey* of context—how uncertainty evolves across a paragraph, how ideas flow and transform, how coherence emerges from local transitions. While working on sentence drift and punctuation topology, I recognized these features capture snapshots of structure, not the continuous texture of narrative progression.

The insight: What if we could visualize this context journey spatially? Perplexity sequences—a language model’s token-by-token uncertainty—encode how surprising each word is given prior context. Human writing, with creative leaps and stylistic variation, should produce complex perplexity landscapes. AI writing, constrained by training distributions, should generate smoother, more predictable patterns.

Convolutional Neural Networks (CNNs) excel at texture classification in computer vision—detecting edges, gradients, and spatial patterns. Recently learning about CNNs in my Information Security course, I realized: *filters and kernels capture contextual structure through local pattern detection, not just sequential decision-making*. Could we apply this to language by converting perplexity sequences into 2D spatial representations?

11.2 Proposed Approach: From Sequence to Texture

11.2.1 Perplexity as a Fingerprint

Given a language model M (e.g., GPT-2) and text sequence $T = \{w_1, w_2, \dots, w_n\}$, perplexity at position i is:

$$\text{PPL}(w_i | w_{<i}) = 2^{-\log_2 P_M(w_i | w_{<i})} \quad (2)$$

A paragraph yields a perplexity sequence $\text{PPL}_T = [\text{PPL}_1, \text{PPL}_2, \dots, \text{PPL}_n]$ with key properties:

- **Low perplexity** (≈ 1): Model confidently predicts next token (common phrases: “the cat sat on the...”)
- **High perplexity** ($\gg 10$): Model surprised by token choice (creative metaphors, domain jargon, unexpected turns)
- **Variance**: Human writing exhibits higher $\sigma^2(\text{PPL}_T)$ due to stylistic variation
- **Smoothness**: AI writing shows autocorrelated perplexity—low drift between consecutive tokens

11.2.2 Spatial Encoding Pipeline

Transformation: Convert 1D perplexity sequences into 2D heatmaps suitable for CNN input:

1. **Perplexity extraction:** Load pre-trained GPT-2 (124M parameters), compute PPL_T via forward pass
2. **Fixed-length normalization:** Pad/truncate to 256 tokens (captures typical paragraph)
3. **2D reshaping:** Reshape $\text{PPL}_T \in \mathbb{R}^{256}$ into $\text{PPL}_{2D} \in \mathbb{R}^{16 \times 16}$ via row-major order
4. **Intensity mapping:** Min-max normalize to $[0, 1]$, map to 8-bit grayscale $[0, 255]$

Visual interpretation:

- **Human text:** High-contrast, speckled texture—frequent perplexity spikes (creative choices) interspersed with smooth regions (grammatical scaffolding). Analogous to natural textures like tree bark or fabric weave.
- **AI text:** Low-contrast, homogeneous texture—gradual perplexity variations, fewer sharp transitions. Analogous to synthetic patterns like machine-printed gradients.

11.2.3 Why CNNs? Capturing Contextual Structure

Traditional NLP features operate on:

- **Bag-of-words:** Discards position (no context journey)
- **N-grams:** Fixed local windows (misses long-range patterns)
- **LSTMs:** Sequential processing (no spatial patterns)
- **Transformers:** Token relationships (not spatial textures)

CNNs offer orthogonal advantages:

- **Local feature extraction:** 3×3 kernels detect perplexity gradients, edges (abrupt transitions), repetitive patterns
- **Translation invariance:** Same texture pattern recognized regardless of position in 16×16 grid
- **Hierarchical learning:** Early layers learn low-level textures (edges, gradients), deeper layers combine into high-level patterns (banding, speckle density)

The context journey emerges spatially: smooth regions indicate predictable flow, sharp edges mark creative pivots, periodic patterns reveal rhythmic structure.

11.3 Proposed CNN Architecture

Lightweight design optimized for small image size (16×16):

Input: (1, 16, 16) grayscale heatmap

Layer 1: Conv2D(in=1, out=32, kernel=3, stride=1, padding=1)
→ ReLU → MaxPool2D(2×2) → (32, 8, 8)

Layer 2: Conv2D(in=32, out=64, kernel=3, stride=1, padding=1)
→ ReLU → MaxPool2D(2×2) → (64, 4, 4)

Layer 3: Conv2D(in=64, out=128, kernel=3, stride=1, padding=1)
→ ReLU → AdaptiveAvgPool2D(1×1) → (128, 1, 1)

Layer 4: Flatten → Linear(128 → 64) → ReLU → Dropout(0.5)

Output: Linear(64 → 2) → Softmax (Human vs AI)

Total Parameters: ~150K

Design rationale:

- 3 convolutional blocks sufficient for 16×16 input
- Channel progression (32→64→128) follows standard pyramid structure
- AdaptiveAvgPool provides global texture summary
- Aggressive dropout (0.5) regularizes against small dataset overfitting

11.4 Expected Benefits and Validation Strategy

11.4.1 Anticipated Advantages

- **Visual interpretability:** Perplexity heatmaps provide intuitive visualization of context flow
- **Fast inference:** CNN forward pass < 1ms (vs. 8ms for DistilBERT)
- **Complementary signal:** Captures temporal uncertainty dynamics orthogonal to embedding-based methods
- **Model collapse detection:** Early warning system for recursive generation degradation

11.4.2 Recursive Generation Experiment (Validation Testbed)

Hypothesis: If perplexity texture degrades visually across recursive generations, CNNs could provide early warning of model collapse before statistical tests detect it.

Protocol:

1. **Gen-0 (Human):** 500 Victorian paragraphs → Perplexity heatmaps → Label: Class 0
2. **Gen-1 (First AI):** Gemini generates on Gen-0 topics → Heatmaps → Label: Class 1
3. **Gen-2 (AI-on-AI):** Fine-tune GPT-2 on Gen-1, generate 500 samples → Label: Class 2
4. **Gen-3-4:** Repeat recursively (train on previous generation)
5. **Total dataset:** 2,500 heatmaps across 5 classes

Expected texture evolution:

- Gen-0 (Human): $\sigma^2 \approx 0.24$, speckled texture, frequent bright spikes
- Gen-1 (AI): $\sigma^2 \approx 0.18$, smoother gradients, fewer extremes
- Gen-2 (AI-on-AI): $\sigma^2 \approx 0.12$, homogenization, horizontal banding
- Gen-3-4: $\sigma^2 \approx 0.08$, near-uniform gray (mode collapse)

Collapse metric:

$$\text{Collapse Score} = \frac{\sigma^2(\text{Gen-0}) - \sigma^2(\text{Gen-N})}{\sigma^2(\text{Gen-0})} \times 100\% \quad (3)$$

Expected: 68% variance reduction from Gen-0 to Gen-4, indicating severe degradation.

11.5 Key Challenges and Mitigation Strategies

11.5.1 Spatial Arrangement Artifacts

Problem: Row-major reshaping ($256 \rightarrow 16 \times 16$) creates artificial boundaries—token 16 and token 17 are sequential but appear spatially distant (end of row 1, start of row 2).

Mitigation strategies:

- **Hilbert curve mapping:** Space-filling curve preserves locality better than row-major
- **Overlap stitching:** Reshape with overlapping windows (e.g., 20×20 grid from 256 tokens via sliding window)
- **Circular padding:** Treat image as toroidal (top row wraps to bottom) to eliminate edge effects

11.5.2 Perplexity Model Selection

Problem: Using GPT-2 to evaluate GPT-4 outputs may yield non-informative heatmaps (uniformly high perplexity).

Mitigation:

- Use matched models: Evaluate GPT-3 outputs with GPT-3 perplexity
- Z-score standardization within-model before cross-model comparison
- Ensemble: Generate heatmaps from multiple LMs (GPT-2, BERT, LLaMA), concatenate as 3-channel RGB image

11.5.3 Small Dataset Overfitting

Problem: 3,492 samples insufficient for training CNNs from scratch (ImageNet has millions).

Mitigation:

- **Transfer learning:** Pre-train CNN on texture benchmark (DTD: 5,640 images), fine-tune on perplexity heatmaps
- **Data augmentation:** Rotate 90°, flip horizontally (preserves texture, increases dataset 4×)
- **Aggressive regularization:** L2 penalty, dropout 0.5, early stopping

11.6 Relation to Current Work: Four-Tier Ensemble

This approach would complement the existing detection hierarchy:

Tier	Method	Signal Captured	Inference
A	XGBoost + Features	Syntactic substrate	0.8ms
B	GloVe + NN	Semantic content	1ms
C	DistilBERT + LoRA	Contextual patterns	8ms
D (Proposed)	CNN + Perplexity	Uncertainty dynamics	1ms

Table 19: Multi-tier detection system with proposed CNN extension

An ensemble voting mechanism (weighted by tier accuracy) could achieve robustness across different attack vectors:

- Blind mutations disrupt statistical features → Tier A fails, Tiers B-D catch
- Synonym replacement alters semantics → Tier B fails, Tiers A,C,D catch
- Victorian-style prompting mimics syntax → Tier C struggles, Tier D detects perplexity smoothness

11.7 Implementation Timeline (If Pursued)

Phase 1: Feasibility validation

- Generate perplexity heatmaps for existing 3,492-sample dataset using GPT-2
- Manual visual inspection: Are human vs. AI textures distinguishable?
- Compute variance degradation across recursive generations (if data available)
- *Decision point:* Proceed only if visual separation is clear

Phase 2: Baseline implementation

- Implement lightweight CNN (3 conv layers, 150K parameters) in PyTorch
- Transfer learning from texture classification benchmark (Describable Textures Dataset)
- Train on human vs. AI binary classification
- Baseline performance: Target 85-90% accuracy

Phase 3: Rigorous evaluation

- Adversarial robustness testing (does back-translation alter perplexity texture?)

- Comparison with existing perplexity-based detectors (GPTZero’s burstiness score)
- Multi-model perplexity ensembles (GPT-2, BERT, LLaMA as 3-channel input)
- Ablation studies on spatial encoding methods (row-major vs. Hilbert curve)

11.8 Why This Remains Conceptual

Due to time constraints and competing priorities (ICSA 2026 workshop deadline, Literary Club co-ordination), I did not implement this approach. The core ideas emerged during feature engineering when exploring perplexity as a discriminator, but translating this to a CNN architecture requires:

1. **Careful spatial encoding design** to avoid artifacts from arbitrary reshaping
2. **Extensive experimentation** with baseline choices (padding tokens vs. zero-vectors vs. mean embeddings)
3. **Validation that texture patterns generalize** across domains (Victorian literature is a narrow testbed)

These considerations deserve dedicated attention rather than rushed implementation. The approach represents genuine research potential but requires proper experimental validation.

11.9 Comparison to Existing Perplexity Methods

Method	Representation	Classifier	Granularity
GPTZero	Scalar (mean PPL)	Threshold	Document-level
Originality.ai	Burstiness (PPL variance)	Threshold	Paragraph-level
Proposed	2D heatmap (PPL texture)	CNN	Token-level spatial

Table 20: Perplexity utilization: Proposed approach vs. existing tools

Key differentiators:

- **Spatial encoding:** Existing tools collapse sequence to scalars—lose positional information. Proposed approach preserves local structure via 2D mapping.
- **Learnable features:** Thresholds require manual tuning per domain. CNN learns discriminative textures from data.
- **Multi-class capability:** Binary (human/AI) vs. proposed 5-class recursive generation tracking.

11.10 Broader Implications

If successful, this methodology generalizes beyond AI detection to any NLP problem with sequential uncertainty patterns:

- **Machine translation quality:** Perplexity texture could reveal unnatural phrasing before BLEU scores detect it
- **Language model calibration:** Heatmaps visualize confidence distributions across contexts
- **Text coherence evaluation:** Texture smoothness correlates with logical flow
- **Authorship attribution:** Each author may have unique “perplexity signature” (vocabulary diversity patterns)

The cross-modal paradigm—applying computer vision to language modeling artifacts—opens research directions at the intersection of NLP and CV. This project provides the conceptual blueprint; future work would execute the vision with rigorous experimentation.

11.11 Conclusion

Cross-modal perplexity texture analysis represents a promising research direction inspired by the challenges encountered during this project’s feature engineering phase. The core insight—that context evolves spatially and CNNs can capture this journey through texture classification—deserves proper investigation.

Key contributions of this proposal:

- Novel problem framing (perplexity as 2D texture, not 1D sequence)
- Concrete architecture design (lightweight CNN for small images)
- Validation strategy (recursive generation experiment)
- Honest assessment of challenges (spatial artifacts, model selection, overfitting)

Next steps (if pursued):

1. Visual feasibility check: Generate heatmaps, confirm human vs. AI separation
2. Baseline CNN implementation with transfer learning
3. Ablation studies on encoding methods
4. Comparison with scalar perplexity approaches

References

- [1] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). LoRA: Low-Rank Adaptation of Large Language Models. *arXiv preprint arXiv:2106.09685*.
- [2] Sundararajan, M., Taly, A., & Yan, Q. (2017). Axiomatic Attribution for Deep Networks. *Proceedings of the International Conference on Machine Learning (ICML)*, 3319–3328.
- [3] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is All You Need. *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, 5998–6008.
- [4] Darmon, A. N. M., Bazzi, M., Howison, S. D., & Porter, M. A. (2020). Pull out all the stops: Textual analysis via punctuation sequences. *European Journal of Applied Mathematics*, 32(6), 1069–1105.
- [5] Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum Associates.
- [6] Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.
- [7] Khmelev, D. V., & Tweedie, F. J. (2001). Using Markov Chains for Identification of Writers. *Literary and Linguistic Computing*, 16(4), 299–307.
- [8] Amancio, D. R. (2015). A Complex Network Approach to Stylometry. *PLOS ONE*, 10(8), e0136076.
- [9] Levenshtein, V. I. (1966). Binary codes capable of correcting deletions, insertions, and reversals. *Soviet Physics Doklady*, 10(8), 707–710.
- [10] Pennington, J., Socher, R., & Manning, C. D. (2014). GloVe: Global Vectors for Word Representation. *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 1532–1543.
- [11] Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.
- [12] Alzantot, M., Sharma, Y., Elgohary, A., Ho, B. J., Srivastava, M., & Chang, K. W. (2018). Generating Natural Language Adversarial Examples. *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2890–2896.
- [13] Jin, D., Jin, Z., Zhou, J. T., & Szolovits, P. (2020). Is BERT Really Robust? A Strong Baseline for Natural Language Attack on Text Classification and Entailment. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(05), 8018–8025.
- [14] Kirchenbauer, J., Geiping, J., Wen, Y., Katz, J., Miers, I., & Goldstein, T. (2023). A Watermark for Large Language Models. *Proceedings of the International Conference on Machine Learning (ICML)*, 17061–17084.